

Thm 1.11 If $f: [a, b] \rightarrow \mathbb{R}$ is continuous then f is Riemann integrable.

Proof Since f is continuous on a closed, bounded interval, then f is bounded and uniformly continuous.

let $\varepsilon > 0$ and choose $\delta > 0$ such that

$$s, t \in [a, b], |s - t| < \delta \Rightarrow |f(s) - f(t)| < \varepsilon. \quad (*)$$

let $n \in \mathbb{N}$ be such that $\frac{b-a}{n} < \delta$, and choose an equally spaced partition $P = \{x_0 = a, \dots, x_n = b\}$ with

$$x_j - x_{j-1} = \frac{b-a}{n}, \quad j = 1, \dots, n.$$

Then

$$\begin{aligned} U(f, [a, b]) - L(f, [a, b]) &\leq U(f, P, [a, b]) - L(f, P, [a, b]) \\ &= \frac{b-a}{n} \sum_{i=1}^n (\sup_{[x_{j-1}, x_j]} f - \inf_{[x_{j-1}, x_j]} f) \\ &\stackrel{(*)}{\leq} \frac{b-a}{n} n \varepsilon = (b-a) \cdot \varepsilon. \end{aligned}$$

letting $\varepsilon \rightarrow 0$ we conclude that $U(f, [a, b]) \leq L(f, [a, b])$, hence $\Rightarrow$.

Why the Riemann integral is not good enough...

- Riemann integration does not handle functions with many discontinuities
- Riemann integration does not handle unbounded functions
- Riemann integration does not work well with limits:

$$f_n \rightarrow f \text{ (in some sense)} \stackrel{?}{\Rightarrow} \int_a^b f_n \rightarrow \int_a^b f$$

Example 1.12 [Discontinuous function]

$$f: [0,1] \rightarrow \mathbb{R}, \quad f(x) := \begin{cases} 1 & x \in \mathbb{Q}, \\ 0 & x \in [0,1] \setminus \mathbb{Q}. \end{cases}$$

If $[a,b] \subset [0,1]$, $a < b$, then $\inf_{[a,b]} f = 0$, $\sup_{[a,b]} f = 1$

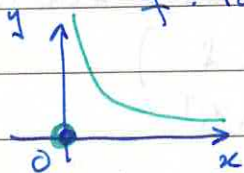
Thus $L(f, P, [0,1]) = 0$, $U(f, P, [0,1]) = 1 \quad \forall P \text{ partition}$,
hence $L(f, [0,1]) = 0 < U(f, [0,1]) = 1$.

We conclude that f is not Riemann integrable.

(See Example 9.3* ... it is Lebesgue integrable)

Example 1.13 [Unbounded functions]

$$f: [0,1] \rightarrow \mathbb{R}, \quad f(x) := \begin{cases} \frac{1}{\sqrt{x}} & \text{if } 0 < x \leq 1, \\ 0 & \text{if } x = 0. \end{cases}$$

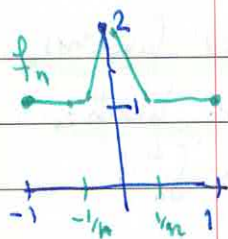


Let P be a partition of $[0,1]$. Then $\sup_{[x_0, x_1]} f = +\infty$
 $\lim_{x \rightarrow 0} \frac{1}{\sqrt{x}} = +\infty$

$$\Rightarrow U(f, P, [0,1]) = +\infty.$$

However, we should consider the area under the graph of f to be 2 and not ∞ , because

$$\lim_{a \downarrow 0} \int_a^1 f = \lim_{a \downarrow 0} \left[2\sqrt{x} \right]_a^1 = \lim_{a \downarrow 0} (2 - 2\sqrt{a}) = 2.$$



$f_n \rightarrow f$ point.
not unif.

$$f(x) = \begin{cases} 1 & x \neq 0 \\ 2 & x = 0 \end{cases}$$

Def 1.14 Let X be a set, and let $f_n, f: X \rightarrow \mathbb{R}$.

$f_n \rightarrow f$ pointwise if $\forall x \in X \quad \lim_{n \rightarrow \infty} f_n(x) = f(x)$, i.e.
 $\forall x \in X \quad \forall \epsilon > 0 \quad \exists n = n(\epsilon, x) \quad \forall n \geq n_0 \quad |f_n(x) - f(x)| < \epsilon.$

$f_n \rightarrow f$ uniformly if $\forall \epsilon > 0 \quad \exists n_0 = n_0(\epsilon) \quad \forall n \geq n_0 \quad \forall x \in X \quad |f_n(x) - f(x)| < \epsilon.$

Recall: the uniform limit of continuous functions is also continuous

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Example 1.15 [Limits of Riemann Integrable functions]

Let $Q \cap [0,1] = \{r_1, r_2, \dots, r_n, \dots\}$. Let

$$f_k(x) = \begin{cases} 1 & x \in \{r_1, \dots, r_k\}, \\ 0 & \text{---} \end{cases}$$

Then f_k is Riemann integrable, with $\int_0^1 f_k = 0$.
However, $f_k \rightarrow f$ pointwise, $f(x) = \begin{cases} 1 & x \in Q \cap [0,1], \\ 0 & \text{---} \end{cases}$

and this function is not Riemann integrable. (see Example 1.12).

But the following is a positive result:

Proposition 1.16 Let $f_n : [a,b] \rightarrow \mathbb{R}$ be Riemann integrable functions with $|f_n(x)| \leq M \quad \forall n \in \mathbb{N}, x \in [a,b]$.

Let $f : [a,b] \rightarrow \mathbb{R}$ be the pointwise limit,

$$f_n(x) \rightarrow f(x) \quad \forall x \in [a,b]$$

(If) f is Riemann integrable then

$$\int_a^b f = \lim_{n \rightarrow \infty} \int_a^b f_n \quad (\text{interchange limit}).$$

Remark 1.17 Note the (If)!

We'll give a proof later on using measure theory.
Further, Prop 1.16 also on $f_n \rightarrow f$ uniformly.
(don't need the (If)) (see Remark 10.11)

The boundedness of $\{f_n\}$ is important.

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then draw rectangles to approximate area under the curve.

Riemann integral: partition the domain (count sequentially)

Lebesgue integral: partition the range - look @ points with "same value" of f and aggregate them (stack and sort)

Goal:

• $s: X \rightarrow \mathbb{R}$ (simple function) $s = \sum_{i=1}^N a_i \chi_{A_i}$

A_i "nice" sets, $A_i = s^{-1}(\{a_i\})$

$$\int_X s \, d\mu = \sum_{i=1}^N a_i \mu(A_i)$$

↑ need the concept of measure

• $f \geq 0$, define $\int_X f \, d\mu := \sup_{s \leq f} \int_X s \, d\mu$

2. Outer Measures

Consider a set X (e.g., $X = \mathbb{R}^N \times I$ with $I = [a, b] \subseteq \mathbb{R}$).

Goal: to "measure" an arbitrary set $E \subseteq X$.

Idea: approximate E with unions of "nice" sets (elementary sets) to which we know how to assign a measure.

E.g., in $\mathbb{R}^N$ use cubes, rectangles, balls

in $\mathbb{R}$ use intervals (a, b) , $[a, b)$, $(a, b]$, $[a, b]$

2.1 Let $\mathcal{E} \subset \mathcal{P}(X)$ be a family of elementary sets, with $\phi \in \mathcal{E}$ and $X = \bigcup_{n=1}^{\infty} X_n$ for some $X_n \in \mathcal{E}$.
 Let $\rho: \mathcal{E} \rightarrow [0, +\infty]$ s.t. $\rho(\phi) = 0$. (elementary measure)
 Define $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ by

$$\mu^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \rho(E_n) : E_n \in \mathcal{E} \quad \forall n \in \mathbb{N}, E \subseteq \bigcup_{n=1}^{\infty} E_n \right\} \quad (2.1)$$

2.2 Lebesgue Outer Measure

If $I \subseteq \mathbb{R}$ is an interval, define $l(I) := \sup I - \inf I$
 (so that $l(I) = l(\overset{\circ}{I}) = l(\overline{I})$). "p(I)"

A rectangle in $\mathbb{R}^N$ is a set of the form

$$R := I_1 \times \dots \times I_N, \quad I_N \subset \mathbb{R} \text{ intervals, with}$$

$$l(R) := \prod_{i=1}^N l(I_i).$$

N.B. $\phi = \prod_{i=1}^N (a, a)$ so $l(\phi) = 0$

$\forall x \in \mathbb{R}^N \quad l(R) = l(x+R)$ for all rectangle $R \subseteq \mathbb{R}^N$
 (translation invariance)

Lebesgue Outer Measure

let $\mathcal{E}$ be the family of rectangles (elementary sets),
with $\phi \in \mathcal{E}$. let $E \subseteq \mathbb{R}^N$.

Define
 $(\mathcal{L}^N)^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \ell(R_n) : E \subseteq \bigcup_{n=1}^{\infty} R_n, R_n \text{ rectangle} \right\}$

... N-dimensional Lebesgue outer measure

2.3 Lebesgue-Stieltjes Outer Measure

let $I \subseteq \mathbb{R}$ be an open interval and let $f: I \rightarrow \mathbb{R}$ be an increasing function.

let $\mathcal{E}$ be the family of all intervals $(a, b]$, $a, b \in I$,
 $a \leq b$. Set $\mu_f: \mathcal{E} \rightarrow [0, +\infty)$ to be
 $\mu_f((a, b]) := f(b) - f(a) (\geq 0)$.

Given $E \subseteq I$, the Lebesgue-Stieltjes outer measure of
 E generated by μ_f is

$$\mu_{\mu_f}^*(E) := \inf \left\{ \sum_{n=1}^{\infty} \overbrace{(f(b_n) - f(a_n))}^{\mu_f((a_n, b_n])} : a_n, b_n \in I, \right. \\ \left. a_n \leq b_n, E \subseteq \bigcup_{n=1}^{\infty} (a_n, b_n] \right\}.$$

2.4 Hausdorff Outer Measure

let $0 \leq s < +\infty$. For $0 < \delta < +\infty$ consider the family
 $\mathcal{E}_\delta := \{ F \subseteq \mathbb{R}^N : \text{diam } F < \delta \}$.

Here, $\text{diam } F := \sup \{ \|x - y\| : x, y \in F \}$, $\|\cdot\|$ Euclidean norm.

Consider the elementary measure

$$\mu_s(F) := \alpha_s \left(\frac{\text{diam } F}{2} \right)^s, \quad F \in \mathcal{E}_\delta$$

where $\alpha_s > 0$ is a constant.

If $E \subseteq \mathbb{R}^N$ define

$$H_\delta^s(E) := \inf \left\{ \sum_{n=1}^{\infty} \alpha_s \left(\frac{\text{diam } E_n}{2} \right)^s : E_n \in \mathcal{E}_\delta, E \subset \bigcup_{n=1}^{\infty} E_n \right\}.$$

If $s=0$ then sum only over those $E_n \neq \emptyset$, with $\left(\frac{\text{diam } E_n}{2} \right)^0 := 1$ even if $\text{diam } E_n = 0$.

Recognizing that $\delta \mapsto H_\delta^s(E)$ is decreasing (because $\delta' < \delta \Rightarrow \mathcal{E}_{\delta'} \subseteq \mathcal{E}_\delta \Rightarrow \inf_{\delta'} \leq \inf_\delta$) we set

$$H_0^s(E) := \lim_{\delta \rightarrow 0^+} H_\delta^s(E) = \sup_{\delta > 0} H_\delta^s(E)$$

... s -dimensional Hausdorff outer measure of E .

Rmk 2.5 often α_s is taken to be 1.

Here, we take $\alpha_s := \frac{\pi^{s/2}}{\Gamma(s/2 + 1)}$ where Γ is the

Euler-Gamma function

$$\Gamma(t) := \int_0^\infty e^{-x} x^{t-1} dx, \quad 0 < t < \infty.$$

Then $\Gamma(n) = (n-1)!$ $\forall n \in \mathbb{N}$ and

$\int_{\mathbb{R}^N} \chi_{B(0,1)} = \alpha_N = (\int_{\mathbb{R}^N} \chi_{B(0,1)})^* (= \int_{\mathbb{R}^N} \chi_{B(0,1)})$ when we'll define the N -dim. Lebesgue meas.

(See Example 4.10(i))

Def 2.6 [outer Measure]

let X be a nonempty set. A map $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ is an outer measure if

- (i) $\mu^*(\emptyset) = 0$; (ii) [Monotonicity] $E \subseteq F \subseteq X \Rightarrow \mu^*(E) \leq \mu^*(F)$
- (iii) [Countable Subadditive] $\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$ for all $\{E_n\} \subset \mathcal{P}(X)$.

We prove that the function μ^* defined in (2.1) is an outer measure.

Prop. 2.7 let X be a nonempty set, let $\mathcal{E} \subset \mathcal{P}(X)$ s.t. $\emptyset \in \mathcal{E}$ and there exist $X_n \in \mathcal{E}$, $n \in \mathbb{N}$, such that $X = \bigcup_{n=1}^{\infty} X_n$.
let $g: \mathcal{E} \rightarrow [0, +\infty]$ be such that $g(\emptyset) = 0$.
Then μ^* defined in (2.1) is an outer measure, and
 $\mu^*(E) \leq g(E)$ for all $E \in \mathcal{E}$. (2.2)

Proof (i) claim $\mu^*(\emptyset) = 0$.

This is clear since $\emptyset \in \mathcal{E}$ and $g(\emptyset) = 0$.

(ii) let $E \subseteq F \subseteq X$. claim $\mu^*(E) \leq \mu^*(F)$.

let $F \subseteq \bigcup_{n=1}^{\infty} F_n$, $F_n \in \mathcal{E}$. Then $E \subseteq \bigcup_{n=1}^{\infty} F_n$, so
$$\mu^*(E) \leq \sum_{n=1}^{\infty} g(F_n).$$

Taking the infimum over the family $\{F_n\}$, we conclude that
$$\mu^*(E) \leq \mu^*(F).$$

(iii) let $E_n \in \mathcal{E}$, $n \in \mathbb{N}$.

claim $\mu^*\left(\bigcup_{n=1}^{\infty} E_n\right) \leq \sum_{n=1}^{\infty} \mu^*(E_n)$.

Fix $\varepsilon > 0$, and for all $n \in \mathbb{N}$ find $\{E_{n,k}\}_{k \in \mathbb{N}} \subseteq \mathcal{E}$ s.t.

$$\sum_{k=1}^{\infty} g(E_{n,k}) \leq \mu^*(E_n) + \frac{\varepsilon}{2^n}, \quad E_n \subseteq \bigcup_{k=1}^{\infty} E_{n,k}.$$

Then $\bigcup_{n=1}^{\infty} E_n \subseteq \bigcup_{n=1}^{\infty} \bigcup_{k=1}^{\infty} E_{n,k}$, hence

$$\mu^* \left(\bigcup_{n=1}^{\infty} E_n \right) \leq \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} g(E_{n+k})$$

$$\leq \sum_{n=1}^{\infty} \left[\mu^*(E_n) + \varepsilon \right] = \sum_{n=1}^{\infty} \mu^*(E_n) + \varepsilon.$$

let $\varepsilon \rightarrow 0^+$.

Finally, let $E \in \mathcal{E}$. claim $\mu^*(E) \leq g(E)$.

This follows trivially from the fact that $\{E\}$ is a covering of E . $\square$

Corollary 2.8 $(\mathcal{L}^N)^*$, μ_f^* , and H_0^s (defined in 2.2, 2.3, 2.4) are outer measures.

and lect #2

Question: When do we have equality in (2.2)?

E.g., would like to have $(\mathcal{L}^N)^*(R) = \lambda(R)$, $R \subset \mathbb{R}^N$ rectangle.

Proposition 2.9 let X be a nonempty set, let $\mathcal{E} \subseteq \mathcal{P}(X)$ s.t. $\emptyset \in \mathcal{E}$ and there exist $X_n \in \mathcal{E}$, $n \in \mathbb{N}$, s.t. $X = \bigcup_{n=1}^{\infty} X_n$. let $g: \mathcal{E} \rightarrow [0, +\infty]$ s.t. $g(\emptyset) = 0$; and let $\mu^*: \mathcal{P}(X) \rightarrow [0, +\infty]$ be defined as in (2.1).

Then $\mu^*(E) = g(E)$ for every $E \in \mathcal{E}$ iff g is countable subadditive, i.e.,

$$g(E) \leq \sum_{n=1}^{\infty} g(E_n)$$

whenever $E, E_n \in \mathcal{E}$, $n \in \mathbb{N}$, and $E \subseteq \bigcup_{n=1}^{\infty} E_n$.

Proof [$\Leftarrow$] Assume that g is countable subadditive. let $E, E_n \in \mathcal{E}$.

let $E \subseteq \bigcup_{n=1}^{\infty} E_n$, $n \in \mathbb{N}$. Then

$$g(E) \leq \sum_{n=1}^{\infty} g(E_n)$$

and taking the infimum over all such $\{E_n\} \subseteq \mathcal{E}$, we get

$$g(E) \leq \mu^*(E).$$

By (2.2) we have always $\mu^*(E) \leq g(E)$, hence $\mu^*(E) = g(E)$.

$\Rightarrow$ Assume that $\mu^*(E) = g(E)$ for all $E \in \mathcal{E}$. (*)

claim g is countably subadditive.

let $E, E_n \in \mathcal{E}$, $n \in \mathbb{N}$, $E \subseteq \bigcup_{n=1}^{\infty} E_n$.

Then

$$g(E) = \mu^*(E) \leq \sum_{n=1}^{\infty} g(E_n) \quad (\text{by def. of } \mu^*)$$

Remark 2.10

when we introduced the Lebesgue outer measure in $\mathbb{R}^N$, we took as elementary sets rectangles = products of intervals, and intervals could be open, closed, semi-open.

Since $l((a, b)) = l([a, b]) = l([a, b)) = l((a, b])$, the outer measure does not see the difference if we restrict the intervals to be only open, only closed, etc.. Hence, the same applies to rectangles in $\mathbb{R}^N$.

Proposition 2.11

Let R be a rectangle in $\mathbb{R}^N$ (an elementary set) not nec. open or closed.

Then $(\mathcal{L}^N)^*(R) = l(R)$.

Proof: we already know that $(\mathcal{L}^N)^*(R) \leq l(R)$ (see 2.2)

Step: we claim that

$$(\mathcal{L}^N)^*(R) = \inf \left\{ \sum_{n=1}^{\infty} l(R_n) : R \subseteq \bigcup_{n=1}^{\infty} R_n, R_n \text{ open rectangles} \right\} =: \mu^*(R).$$

clearly $(\mathcal{L}^N)^*(R) \leq \mu^*(R)$. (because we are restricting the admissible R 's on $(\mathcal{L}^N)^*$ to obtain μ^*).